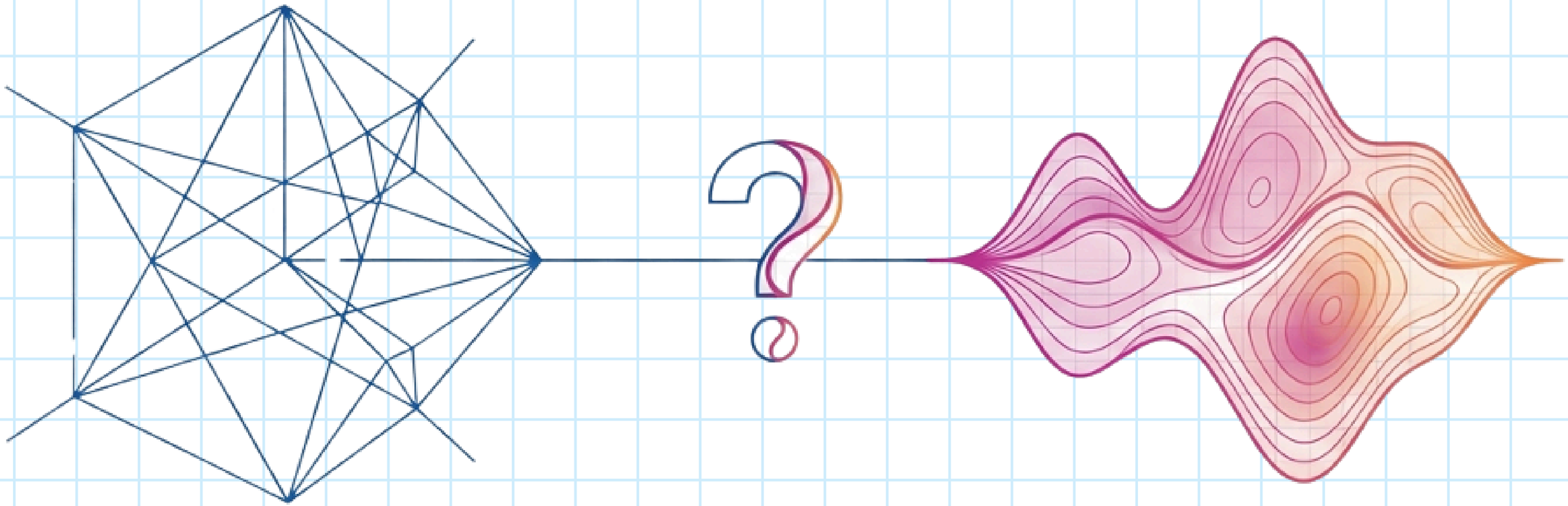
From Linear Scores to a Single Hidden Layer

A Mathematical Study of Simple Learning Systems



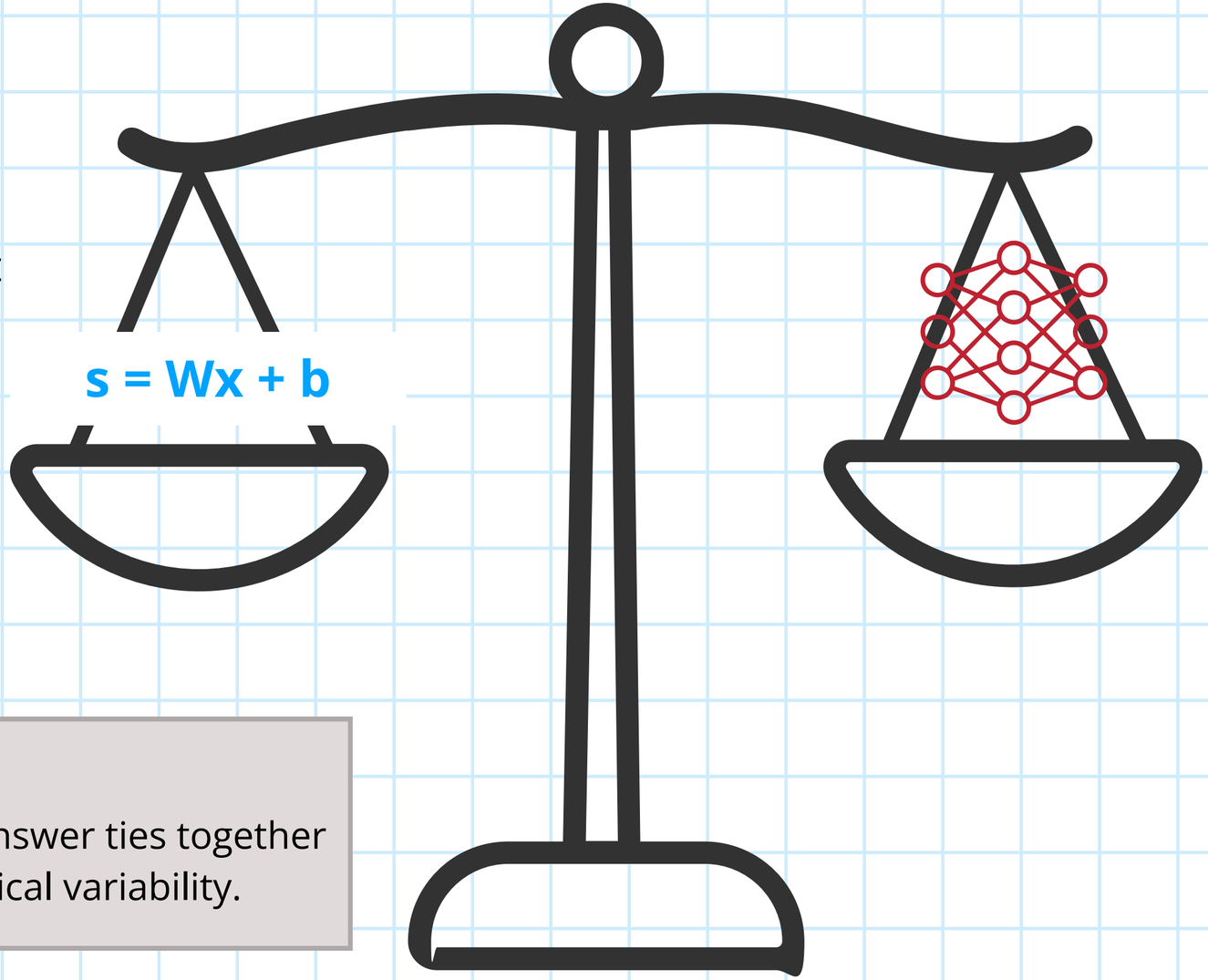
Avaz, Kazim, Raul, and Emil
NAIC - AI Academy

Baku, Azerbaijan
04.04.2026

The Illusion of Default Complexity

Do we always need deep learning?

A prevailing assumption in modern AI is that more parameters automatically yield better results.



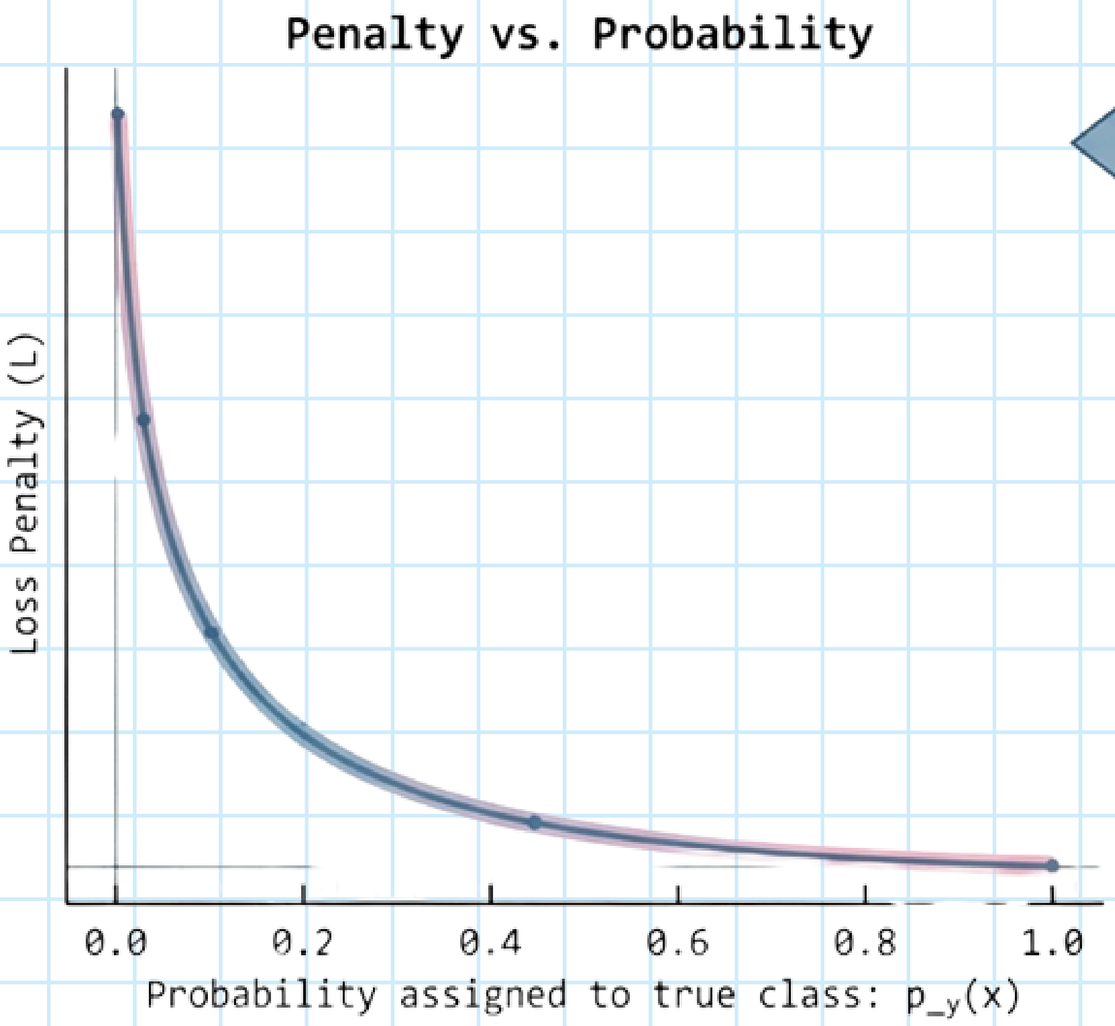
The Core Hypothesis

Bigger models are not automatically better. The answer ties together geometry, the chain rule, optimization, and statistical variability.

The Contenders

	Softmax Regression	One-Hidden-Layer NN
Equations	$s = Wx + b$	$h = \tanh(W_1x + b_1)$ $s = W_2h + b_2$
Core Mechanism	Directly scores the input space via affine transformation.	The tanh function bends and re-encodes the input space before the final linear layer scores it.
Boundary Shape	Convex polytopes (rigid hyperplanes)	Non-linear contours (warped feature geometry)
Complexity	Minimal. No hidden parameters.	Introduces hidden dimensionality (default w=32).

Cross-Entropy as Negative Log-Likelihood



The Objective

$$L(\mathbf{x}, y) = -\log p_y(\mathbf{x})$$

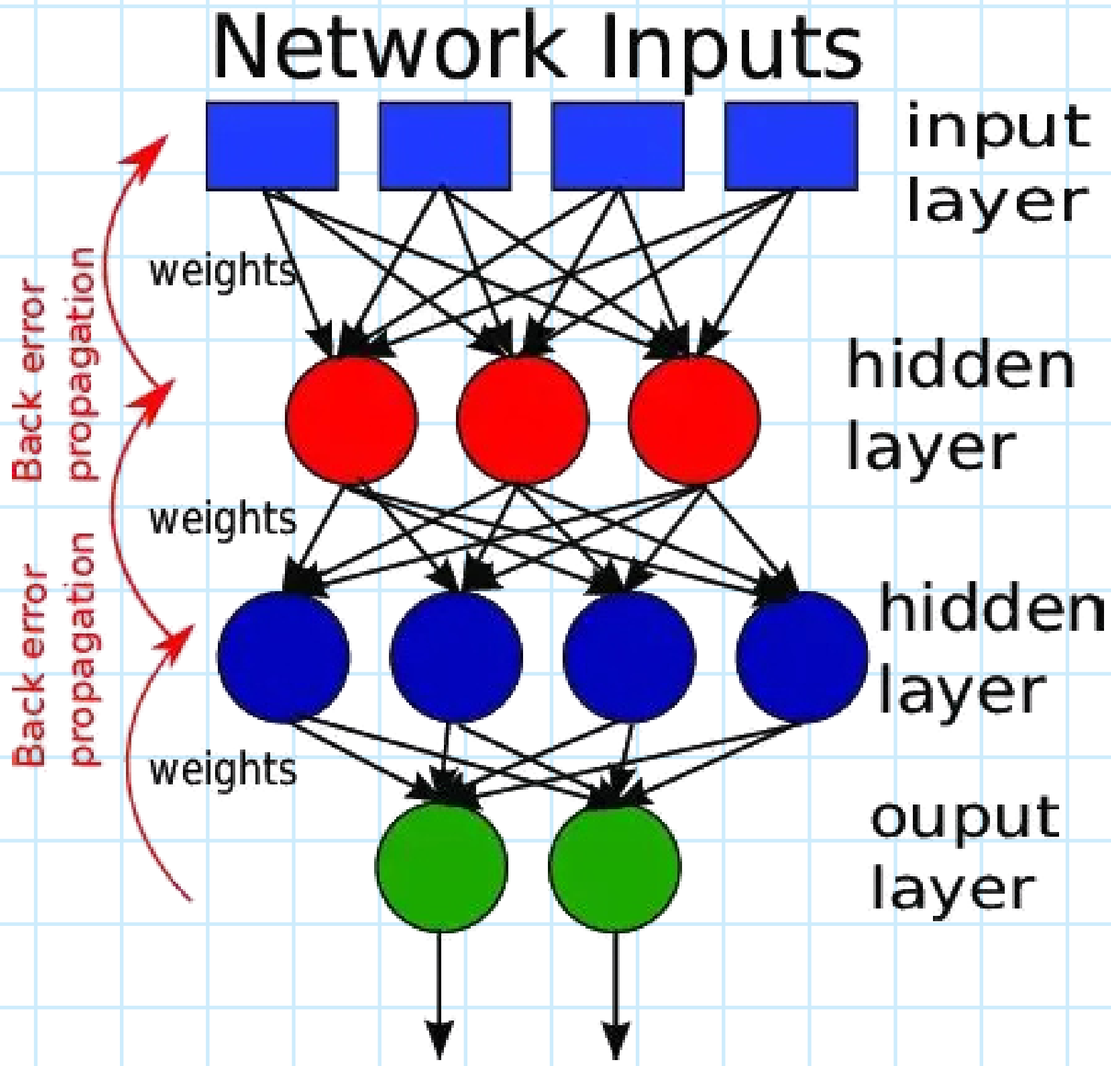
The Mechanism

1. The model outputs a categorical distribution over k classes.
2. The probability it assigns to the true observed label y is exactly $p_y(x)$.
3. Therefore, the log-likelihood of that observation is $\log p_y(x)$.

The Verdict

Averaging this penalty over n independent examples makes minimizing cross-entropy mathematically identical to maximizing likelihood.

Backpropogation



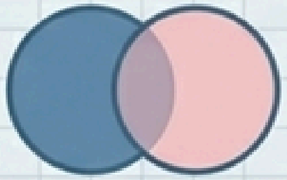
$$dZ^{[L]} = A^{[L]} - Y$$

$$dW^{[l]} = \frac{1}{m} dZ^{[l]} (A^{[l-1]})^T, \quad l = L, L-1, \ldots, 1$$

$$db^{[l]} = \frac{1}{m} \sum dZ^{[l]}$$

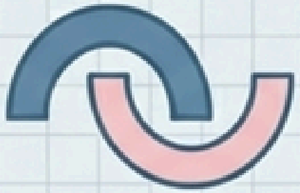
$$dZ^{[l-1]} = (W^{[l]})^T dZ^{[l]} \circ g'^{[l-1]}(Z^{[l-1]}), \quad l = L, L-1, \ldots, 2$$

Experiments



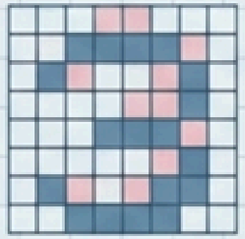
Arena 1: Linear Gaussian

- Two synthetically generated isotropic blobs.
- Hypothesis: Approximately linearly separable.



Arena 2: Interlocking Moons

- 2D configuration where class regions heavily interleave.
- Hypothesis: Requires a nonlinear decision boundary.

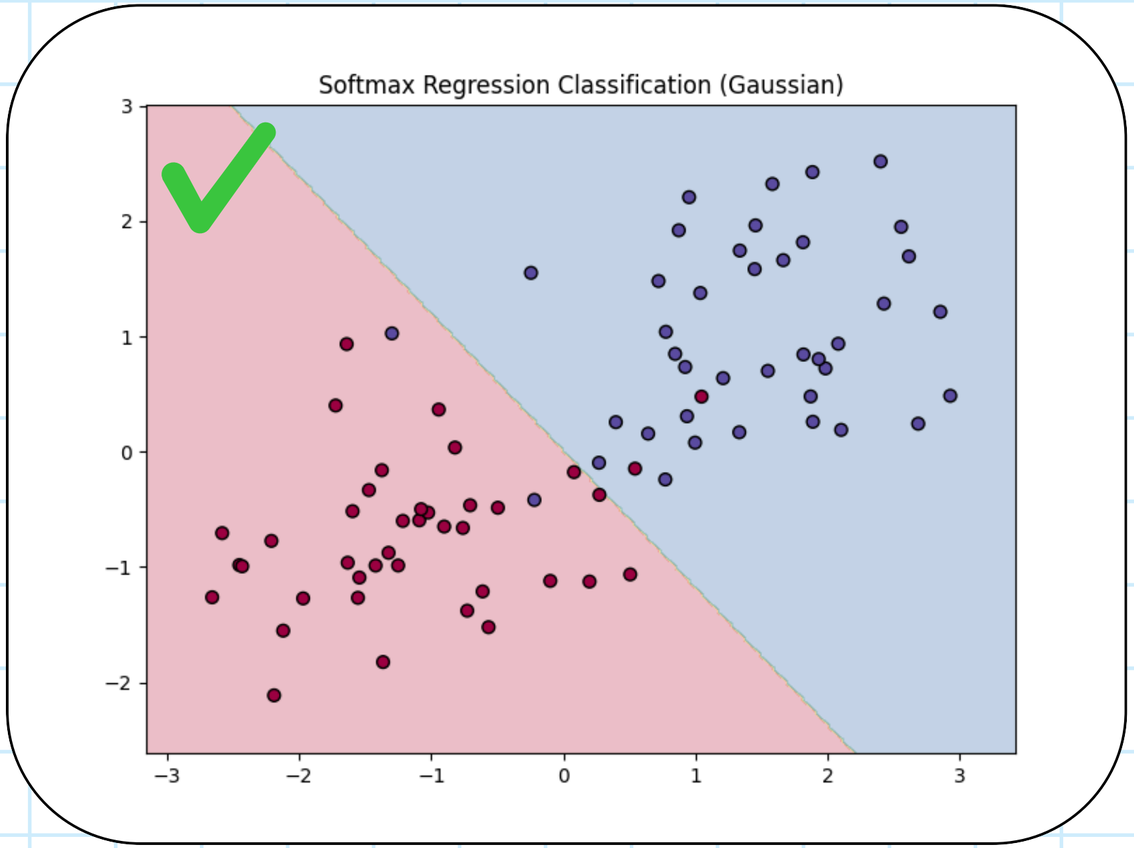


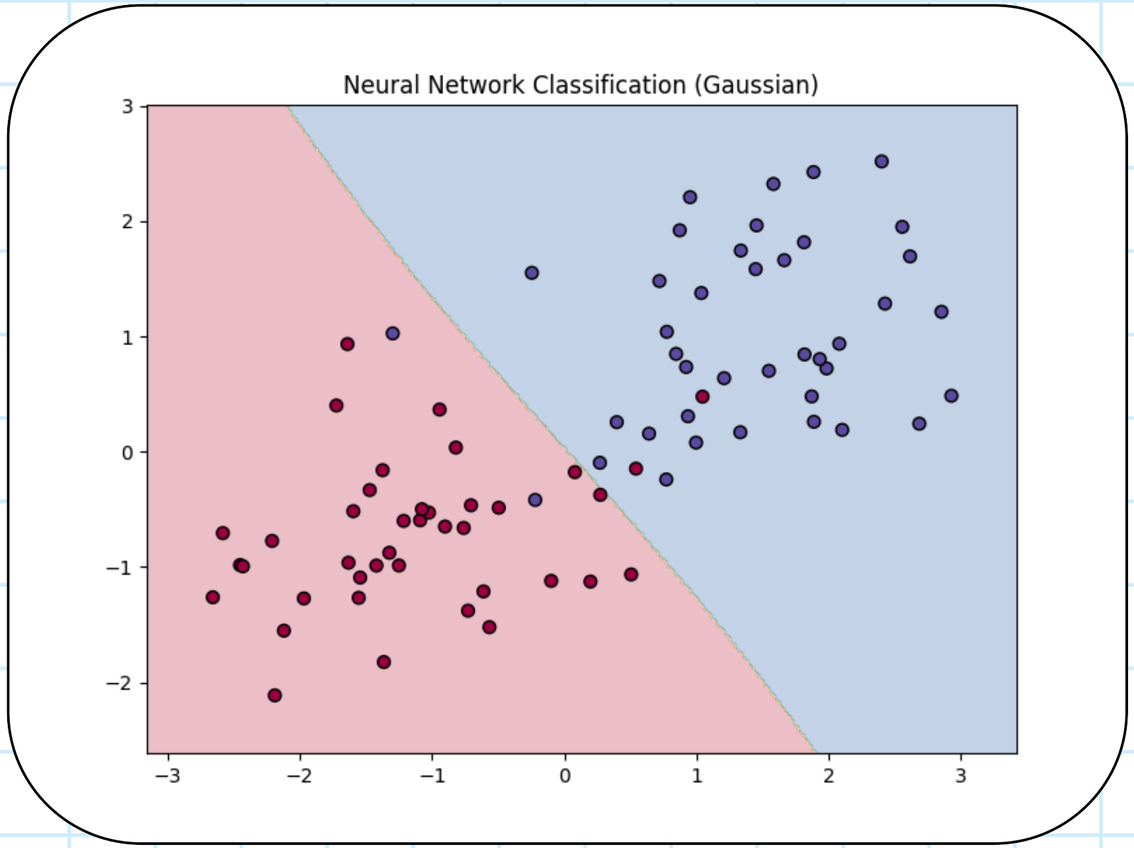
Arena 3: Digits Benchmark

- 8x8 grayscale digit images, flattened to 64-dimensional vectors (k=10).
- Protocol: Strict train/val/test split. Five independent random seeds.

Level 1: The Linear Gaussian Task

Observation: Both architectures **achieve clean separation** of the two clusters.

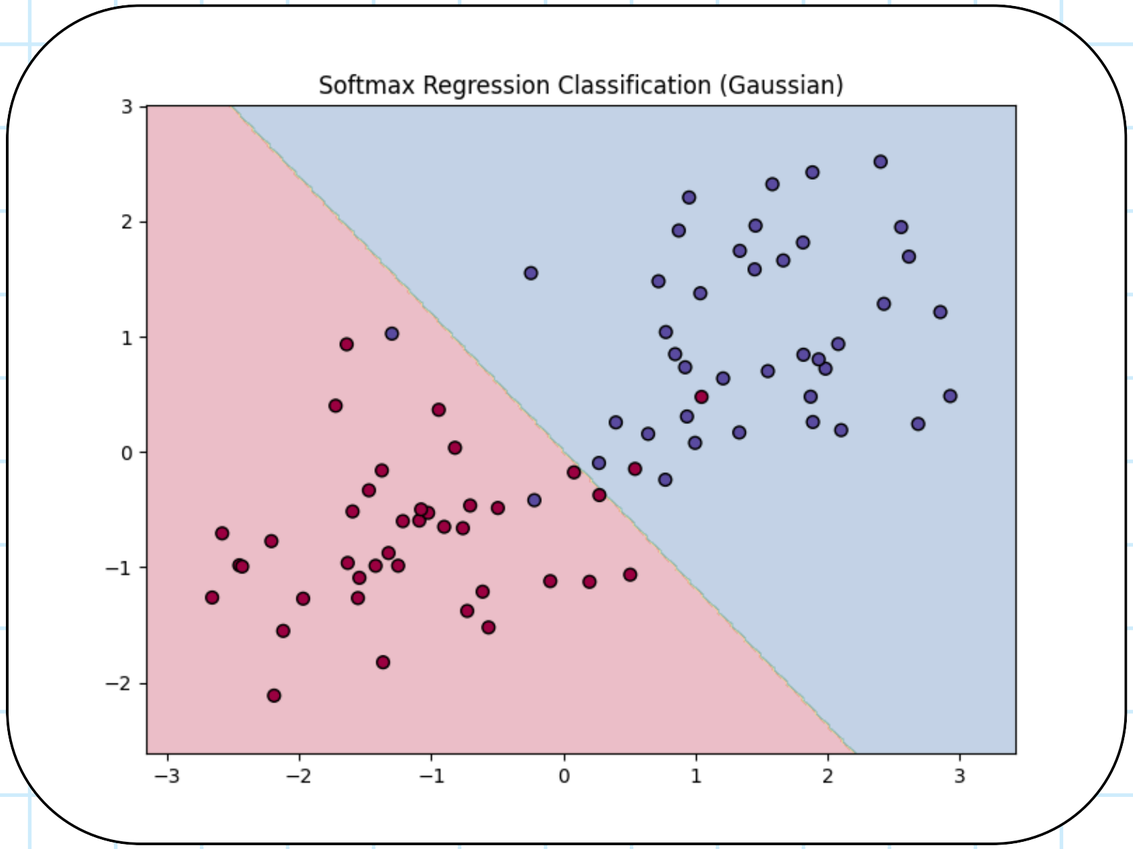


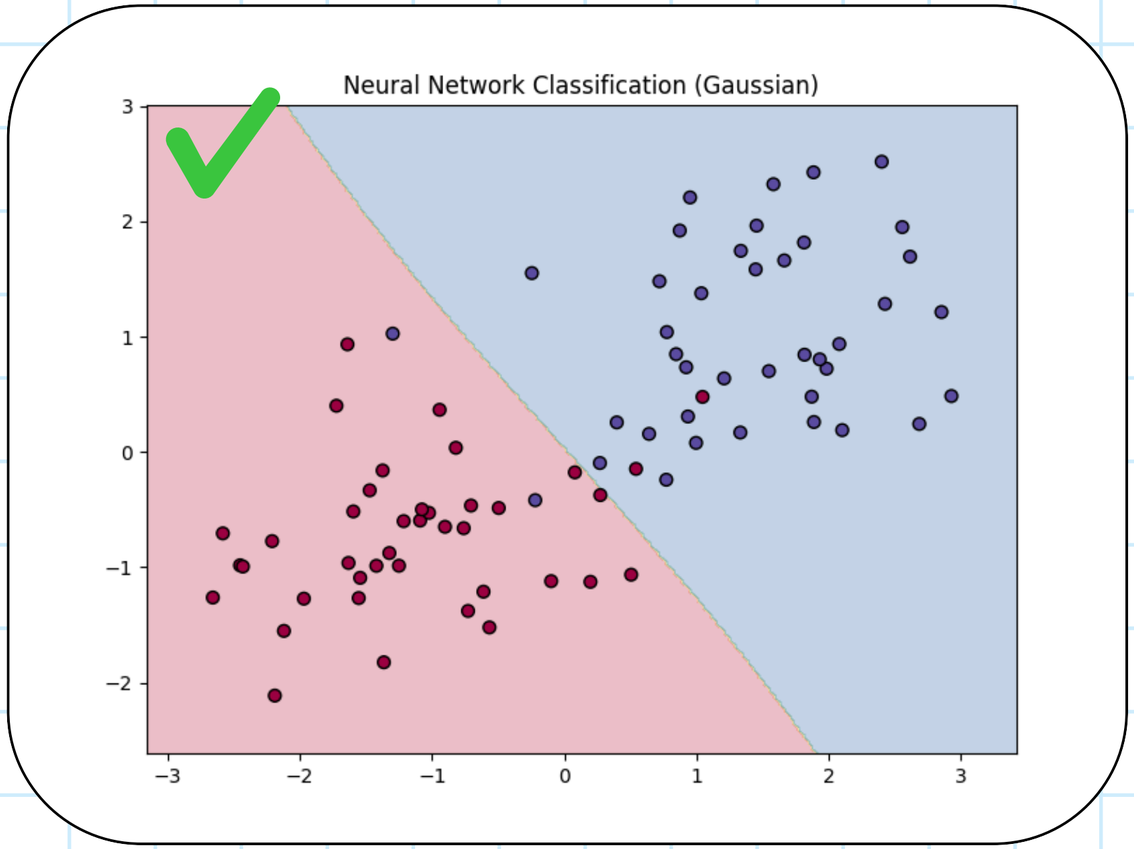


Conclusion:
The linear model's boundary is **sufficient**. The neural network's slightly curved boundary is merely an artifact of its flexibility and offers **no geometric advantage**. Complexity is unnecessary here.

Level 2: The Interlocking Moons Task

Observation: Straight lines can't separate curved data. **Softmax** fails because it is too rigid; the **Neural Network** wins because it is flexible enough to follow the curves.

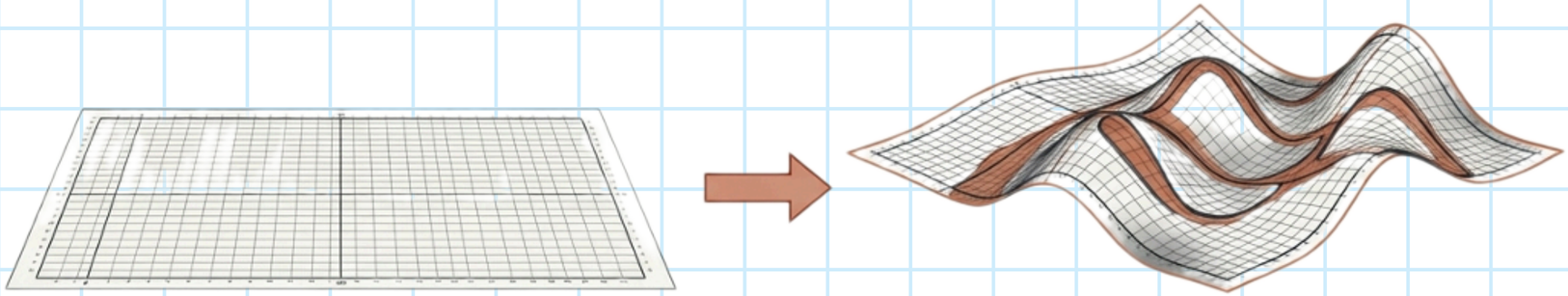


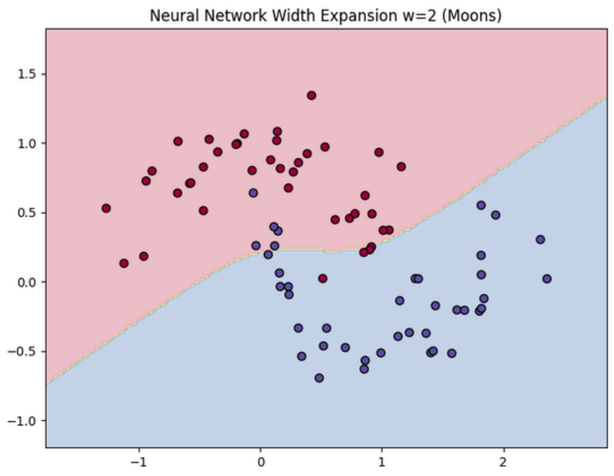


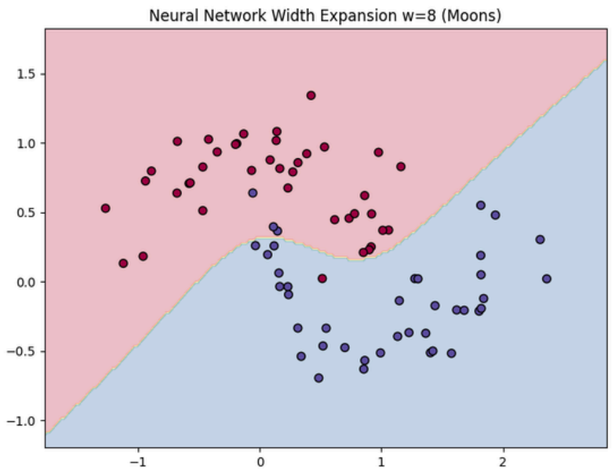
Conclusion:
Structural Failure. Straight lines cannot follow curves. Hidden layers are necessary to handle non-linear patterns.

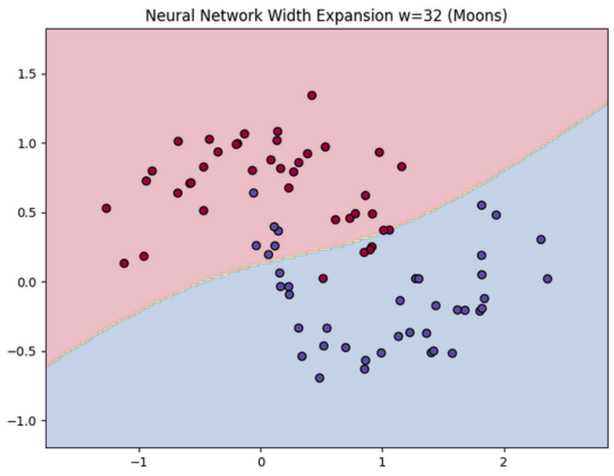
Anatomy of a Non-Linear Fold

What does ‘capacity’ actually look like? Each hidden unit provides ones nonlinear feature fold (**tanh).**









Width 2

Rough, low-detail separation.

Width 8

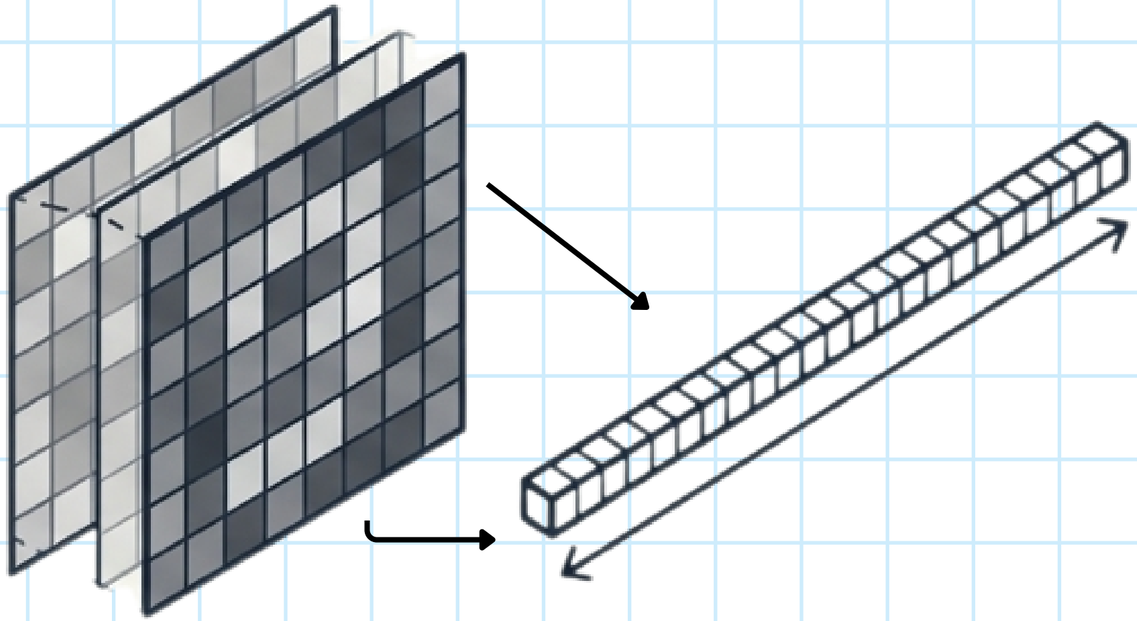
Smooth, well-balanced boundary capturing the shape.

Width 32

Complex, high-precision boundary fitting fine details.

Level 3: The 64-Dimensional Digits Benchmark

Evaluated over **five** independent random initializations to ensure robust, seed-agnostic statistics.



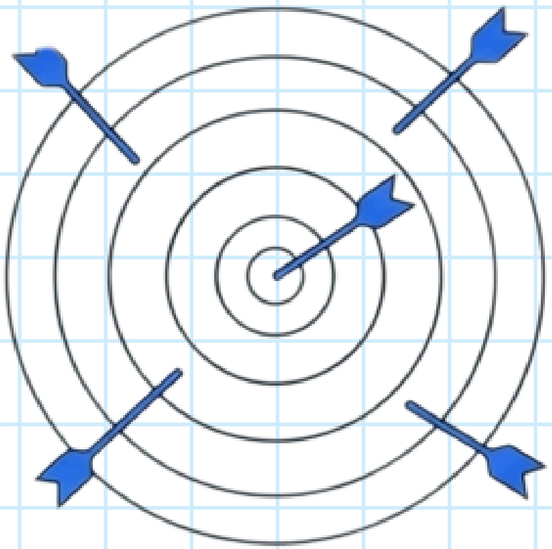
8x8 Input Grid 64-Dim Vector

Metric	Test Accuracy
Softmax Regression	94.73% ± 0.38%
Neural Network	95.22% ± 0.38%

Insight: The accuracy gap is a **mere 0.49** percentage points, and 95% confidence intervals overlap. In terms of decision quality, the two architectures are practically tied.

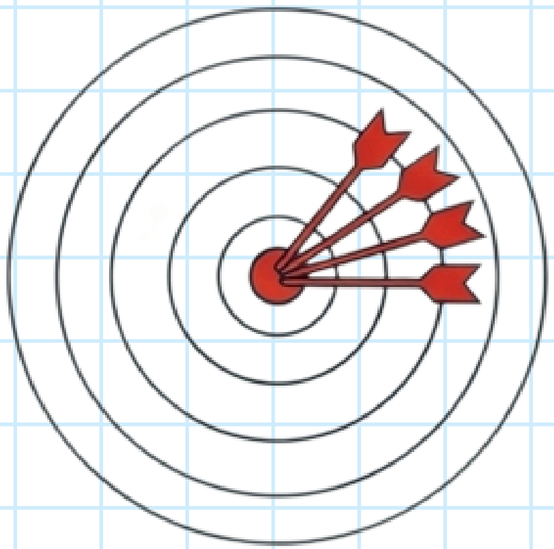
Accuracy vs. Cross-Entropy

Target A: Linear



Linear Cross-Entopy Loss: 0.2315

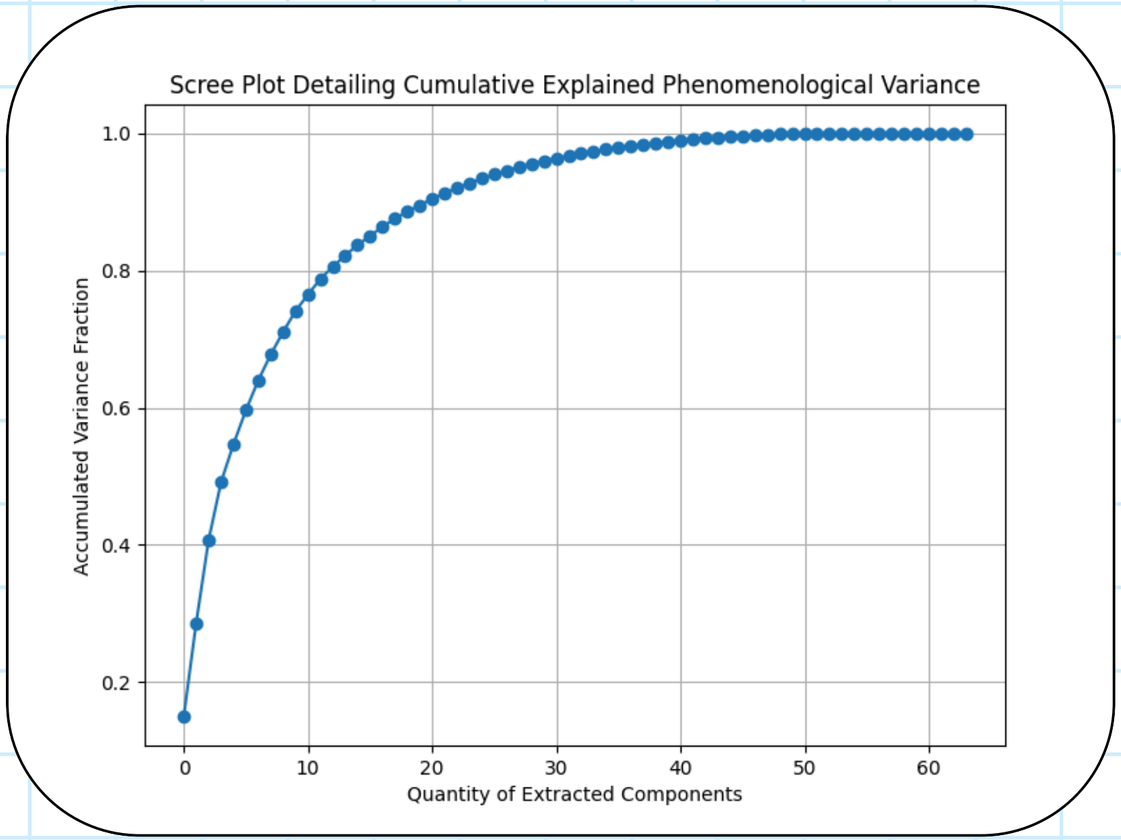
Target B: Neural Net



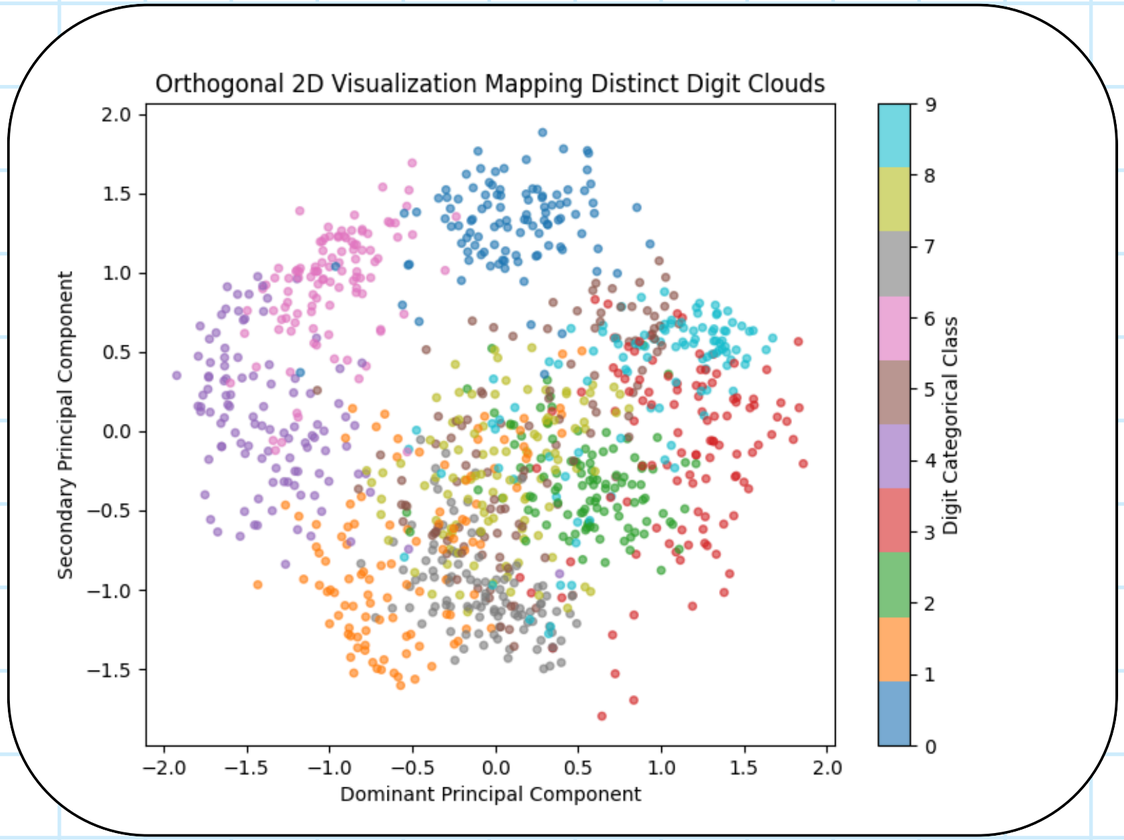
Neural Net Cross-Entropy Loss: 0.1765

Insight: A 23.8% reduction in average loss. While both models make similar discrete predictions, the nonlinear model is substantially better calibrated. It assigns much higher probability to the correct class; crucial for medical, autonomous, or cascading systems where confidence matters.

Track A: PCA/SVD and input geometry



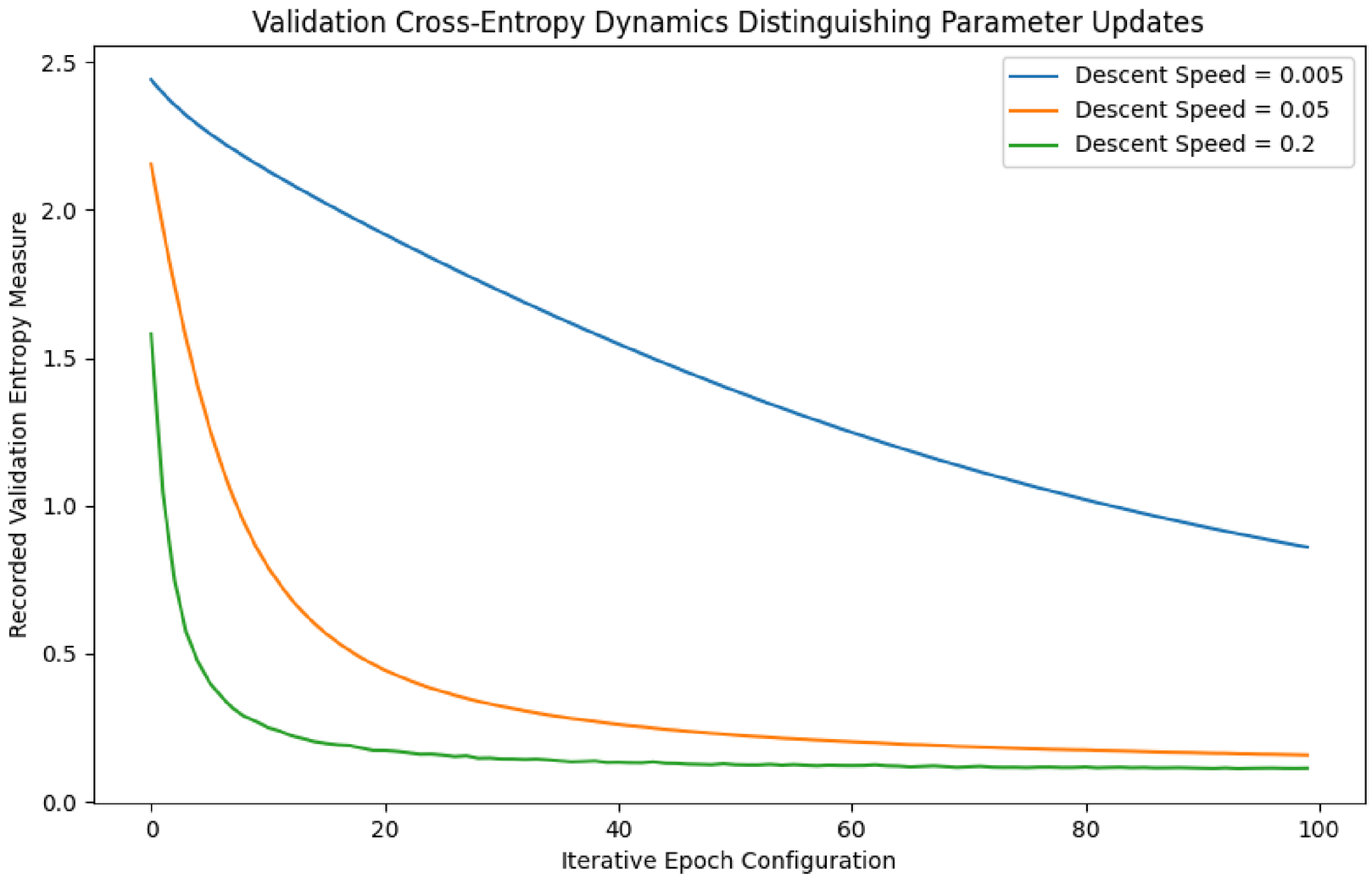
Variance Curve: The plot climbs rapidly, suggesting we don't need all 64 dimension.



2D View: Ortogonal projection reveals class overlap. Two dimensions are insufficient for classification.

Method: PCA analysis via SVD; $X = U\Sigma V^T$ on mean-centered features.

Learning Rate



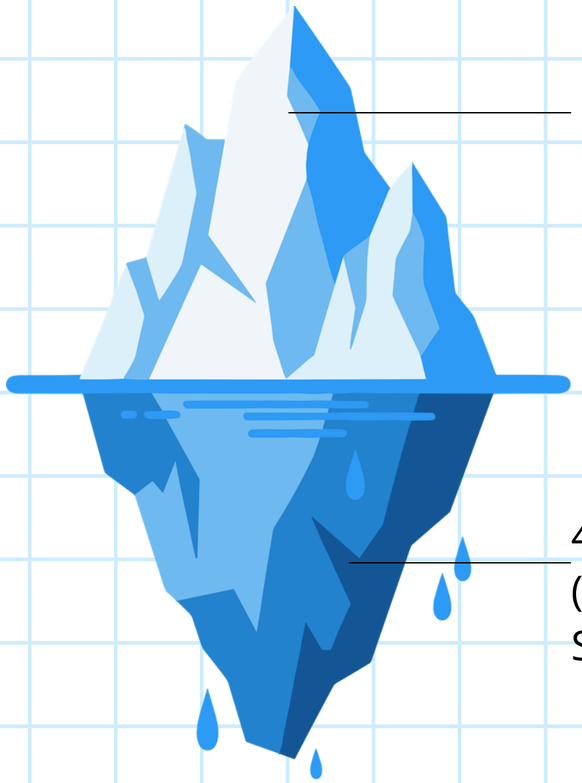
Descent Speed = 0.2: Rapid convergence, stabilizing near 0.1126.

Descent Speed = 0.05: Smooth, steady descent, settling near 0.1571.

Descent Speed = 0.005: Flatlines at 0.8602, essentially stuck at initialization.

Test: Evaluate descent speeds over 100 epochs.
Observation: Validation loss is highly sensitive to step size.
Conclusion: A good architecture fails if the step size is misaligned.

The Low-Rank Linear Subspace



Retraining the Linear model on compressed inputs:	
m=10 components	: 89.95% Accuracy
m=20 components	: 93.21% Accuracy (Loss: 0.2600)
m=40 components	: 94.57% Accuracy (Loss: 0.2316)
m=64 (Full baseline)	: 94.73% Accuracy (Loss: 0.2315)

Conclusion:
We don't need all 64 dimensions to get high accuracy. A smaller **40-dimensional "slice"** contains almost all the useful info. Because the data lives in this flat space, a simple linear model works perfectly.

Synthesis

	Intrinsic Geometry	Linear Result	NN Result
Gaussian Baseline	Linearly Separable.	Clean Separation (Optimal).	Clean Separation (Unnecessary complexity).
Interlocking Moons	Curved / Entangled.	Structural Failure (Rigid bisector)	Smooth Contour (Geometrically essential)
Digits Benchmark	Low-rank linear subspace (64D → 40D).	94.73% Acc (Highly competitive)	95.22% Acc (Superior calibration, 23.8% loss drop)

The Blueprint

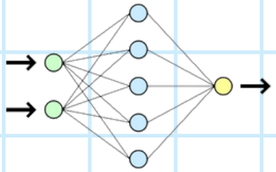
Bigger is not inherently better. Linear structure handles vast amounts of high-dimensional data naturally if the intrinsic rank is low.

Understand the shape of your data before you build the engine. Geometry dictates architecture.

Accuracy is a blunt instrument. Look at cross-entropy to measure true model calibration and confidence.

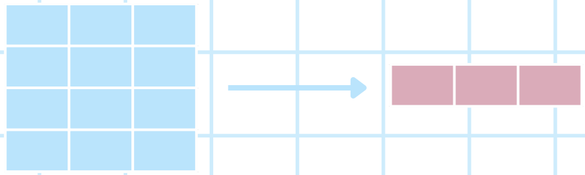
Limitations

Shallow Architecture



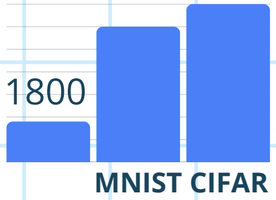
a single **tanh** layer.
DNN with ReLU behaves differently.

No Spatial Structure



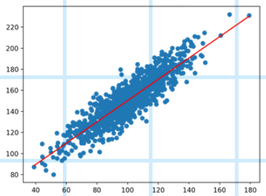
8x8 to 64.
Ignores pixel adjacencies that CNN exploit.

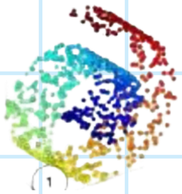
Small Dataset



-**1800** samples.
Full **MNIST/ CIFAR** may perform better.

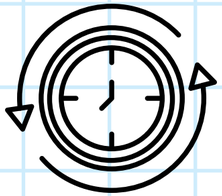
Linear Variance Only





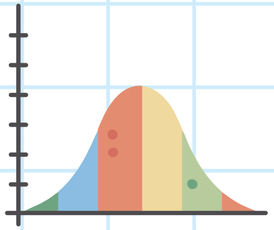
PCA captures linear structure.
UMAP/ t-SNE may tell richer story.

Fixed Optimization



Constant-rate **SDG** used.
Adaptive O. may alter learning-rate sensitivity.

Statistical Sample Size



Only **five** random seeds.
More would tighten the estimates

Thank You!

Any Questions?